

Syllabus

I. Basic Information

Year	2026	Semester	2학기		Class Number	0264
Course Title	경영과학			Class Type	FL	
Class Time	화)12:00~13:30		Classroom	강의동207		
AI Usage Policy for This Course	() Active Use (√) Conditional Use () Prohibited					
	1. Active Use: The use of AI is actively encouraged to support learning. However, students must be aware of the precautions for use and must submit a [Generative AI Usage Statement.] 2. Conditional Use: Limited use of AI is permitted for brainstorming, grammar correction, etc., during learning and assignments. However, the final output must be produced by the student. Students must submit a [Generative AI Usage Statement.] 3. Prohibited: The use of AI is entirely prohibited throughout learning and assignments in this course.					

II. Instructor

Name	김기태	E-mail	kitaekim@kau.ac.kr
Telephone	02-300-0085	Office	본관 324
Office Hours	화14:00-16:00 / 수15:00-17:00 / 목12:00-14:00		
Learning Management System	(√) Y () N () Other Website		
	※ Other Website URL:		

III. Offering Information

Major	Course Code	Course Title	Credits	Course	Grade	Hours per	Lecture	Lab Hours
경영학부	BD4248	경영과학	3	전필	2	3	3	0
경영전공	3BD3206	경영과학	3	전필	2	3	3	0
경영학과	BD4248	경영과학	3	전필	2	3	3	0

IV. Course Description

	(√) ()							
Grading Method	Absolute Grading							
Overview	Management Science is a field of study that uses mathematical modeling and scientific analysis techniques to find the most efficient optimal solutions to complex decision-making problems faced by companies or organizations, and applies these solutions to real-world management to support rational decision-making. In this course, students will learn management science knowledge and problem-solving skills through various methods including theoretical lectures, problem-solving exercises, and hands-on practice.							
Objectives	1) Understanding the necessity of management science and real-world application cases 2) Understanding various methodologies used in management science, such as optimization 3) Acquiring the ability to solve optimization problems using Python 4) Understanding future development directions in management science (research on artificial intelligence and quantum computing in management science)							
Prerequisites	Python							
Type of Lecture	총13주차	01주차, 02주차, 03주차, 04주차, 05주차, 06주차, 07주차, 09주차, 10주차, 11주차, 12주차, 13주차, 14주차						
Teaching Methods								
	() Videos taken from the CTL Studio (√) Self-captured video () Others(Youtube or MOOC) (√) Quiz (√) Submitting Questions () Others		(√) Quiz (√) Lecture (√) Individual Activities () Problem solving by team (√) Discussion (√) Experiment () Others			(√) Assignment for the week (√) Reflective log () Portfolio () Others () None		
Evaluation	Mid-term Exam	Final Exam	Quiz	Assignment	Attendance	Participation	Others	Total
	20	20	20	20	10	10	0	100
Remarks	The participation score includes the reflective logs of students.							

VI. Course Evaluation

Last Course Evaluation Score	/ - / -	Average of Similar Courses Group	/ - / -
Improvement for this Semester			

※ Average of Similar Courses Group : This is the average course evaluation score among all courses within a course group. Course groups are determined based on class sizes, grade and course classification.

VII. Materials and References

Materials	Lecture Slides based on Modern Management Science 9th edition
References	Modern Management Science 9th edition (현대경영과학 (9판), 한빛아카데미, 김세현 저, 2023년 01월 31일 발행)

VIII. Weekly Plan

※ For Captone Design, Weekly Plan might be omitted.

1주차	Objectives	Introduction to Management Science			
	Description	Course introduction, introduction of application cases			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Submitting Questions	60
		수업 중	Lecture	Discussion	75
		수업 후	Reflective log notify	Reflective log	30
	Assignment				
2주차	Objectives	Linear optimization models and linear programming			
	Description	Study linear optimization models and linear programming			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Quiz	60
		수업 중	Quiz Solution	Discussion	75
		수업 후	Reflective log notify	Reflective log	30
	Assignment	Reflective log notify			
3주차	Objectives	Development, application, and limitations of linear optimization models			
	Description	Study development, application, and limitations of linear optimization models			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Quiz	60
		수업 중	Quiz Solution	Discussion	75
		수업 후	Assignment Upload	Assignment	60
	Assignment	Assignment #1			
4주차	Objectives	Analysis of linear optimization models			
	Description	Analysis of linear optimization models using Python on Google Colab			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Submitting Questions	60
		수업 중	Experiment	Individual Activities	75
		수업 후	Reflective log notify	Reflective log	30
	Assignment				

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5주차	Objectives	Post-optimality analysis			
	Description	Study post-optimality analysis			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Quiz	60
		수업 중	Quiz Solution	Discussion	75
		수업 후	Reflective log notify	Reflective log	30
	Assignment				
6주차	Objectives	Sensitivity analysis			
	Description	Study sensitivity analysis			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Quiz	60
		수업 중	Quiz Solution	Discussion	75
		수업 후	Assignment Upload	Assignment	60
	Assignment	Assignment #2			
7주차	Objectives	Analyzing business problems using linear optimization models			
	Description	Study various applications of linear optimization models in real life			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Quiz	60
		수업 중	Quiz Solution	Discussion	75
		수업 후	Reflective log notify	Reflective log	30
	Assignment				
8주차	Objectives	Mid-term Exam			
	Description	Mid-term Exam			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전			
		수업 중			
		수업 후			
	Assignment				

VIII. Weekly Plan

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9주차	Objectives	Integer optimization models			
	Description	Study integer optimization models			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Submitting Questions	60
		수업 중	Lecture	Discussion	75
		수업 후	Reflective log notify	Reflective log	30
	Assignment				
10주차	Objectives	Network models and transportation			
	Description	Study linear and integer optimization models related to networks and transportation			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Submitting Questions	60
		수업 중	Experiment	Individual Activities	75
		수업 후	Assignment Upload	Assignment	30
	Assignment	Assignment #3			
11주차	Objectives	Multi-criteria decision making problems 1: Scoring, DEA			
	Description	Study multi-criteria decision making problems and scoring and DEA methods			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Quiz	60
		수업 중	Quiz Solution	Disucssion	75
		수업 후	Reflective log notify	Reflective log	60
	Assignment				
12주차	Objectives	Multi-criteria decision making problems 2: Goal programming			
	Description	Study the goal programming in multi-criteria decision making problems			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Quiz	60
		수업 중	Quiz Solution	Disucssion	75
		수업 후	Reflective log notify	Reflective log	30
	Assignment				

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13주차	Objectives	Game theory			
	Description	Introduction to game theory			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Quiz	60
		수업 중	Quiz Solution	Disucssion	75
		수업 후	Assignment upload	Assignment	60
	Assignment	Assignment #4			
14주차	Objectives	Future of Management Science			
	Description	Introduction to the future of management science: Agentic AI and Quantum Computing			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전	Self-captured video	Submitting Questions	60
		수업 중	Lecture	Disucssion	75
		수업 후	Reflective log notify	Reflective log	30
	Assignment				
15주차	Objectives	Final Exam			
	Description	Final Exam			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전			
		수업 중			
		수업 후			
	Assignment				
16주차	Objectives	Make-up Class (if necessary)			
	Description	Make-up Class (if necessary)			
	Teaching Methods	단계	교수자	학습자	시간
		수업 전			
		수업 중			
		수업 후			
	Assignment				

IX. Regarding Disabled Students

「Rules Pertaining to Disabled Student Support」 Disabled students are entitled to special conditions or assistance when taking the class, receiving materials, completing homework assignments and writing exams. Students should consult the professor, Academic Affairs Team or Student Affairs Team regarding particular needs prior to registering for the class.

IX. Precautions for Using Generative AI

Generative AI may reflect social and cultural biases embedded in its training data and can produce inaccurate or contextually inappropriate results due to "hallucinations" (factual inaccuracies). Therefore, students must critically review and verify all AI-generated content using reliable sources. If a student fails to recognize these limitations and submits work containing AI-generated errors, the student bears full responsibility for any resulting academic penalties.

Generative AI Usage Statement

1) Basic Information

Course Information	
Student Information	
Tool Used	
Date of Use	

2) Purpose and Scope of Use (check or describe applicable items)

☐	Idea brainstorming and outline drafting
☐	Sentence correction and grammar checking
☐	Data analysis or draft code generation
☐	Requesting summaries or explanations of specific concepts
☐	Other (please specify: _____)
※ Use of generative AI beyond the scope permitted in this course may constitute a breach of academic integrity.	

Main Prompts (Questions)	
Personal Contribution and Revisions	

3) Self-Check and Pledge

☐	I personally verified the factual accuracy of the information generated by AI.
☐	I did not copy the AI's output as-is, but restructured it in my own words.
☐	I used AI only as an auxiliary tool, and derived the key content myself.
☐	I have transparently disclosed my use of generative AI, and understand that failure to do so may be regarded as academic dishonesty.
☐	I recognize that generative AI may produce biased or inaccurate information, and I have critically reviewed its output accordingly.

I confirm that the above statements are true, and that if found to be false, I may face grade penalties or disciplinary actions.

202 년 월 일

Name: _____(Signature)

※ The content of the Generative AI Usage Statement is intended to ensure transparency and is not directly reflected in grading.